

SESSION 1

Imaging Informatics II

Data Acquisition & DICOM Foundations

From the patient and modality to structured digital imaging information

Course	RAD-423 Imaging Informatics
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Suggested study time	60–90 minutes

Learning Objectives

By the end of this session, the student should be able to:

1. define imaging informatics and describe its role in a modern imaging department;
2. explain how analogue or physical signals are converted into digital imaging data;
3. compare the principal kinds of data generated by X-ray, CT, MRI, ultrasound and nuclear medicine;
4. explain the purpose of DICOM in medical imaging communication and information management; and
5. distinguish image pixel data from the patient, study and acquisition information stored with an imaging object.

Key Concept

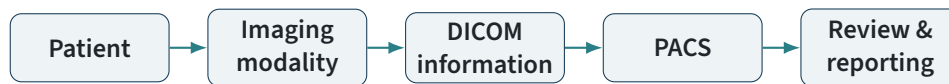
Imaging informatics is the information backbone of modern radiology. It connects the patient, imaging modality, image data, clinical information, storage systems and healthcare professionals so that the correct examination can be acquired, stored, retrieved, interpreted and communicated.

1 Imaging Informatics: The Big Picture

Imaging informatics is the application of information technology, computer science, data management and communication systems to the acquisition, storage, retrieval, distribution, analysis and use of medical images.

In practical terms, an imaging department does not operate as a collection of isolated scanners. A CT or MRI system must receive correct patient and examination information, generate image data, package that information in an agreed format, transfer it to storage, and make it available to clinicians. The informatics system is what allows

these steps to behave as one coordinated workflow.



The simplified chain above hides many additional systems, including the Radiology Information System (RIS), Electronic Health Record (EHR), networks, archives, quality assurance tools, artificial intelligence systems and analytics platforms. Later sessions examine those systems individually.

Clinical / Practical Link

A beautifully acquired CT examination is of little value if it is stored under the wrong patient, cannot reach PACS, or cannot be retrieved by the radiologist. Image quality and information quality are therefore linked.

1.1 The information handled by imaging systems

Medical imaging generates considerably more than pictures. Depending on the examination, data may include:

- patient demographics and identifiers;
- examination and scheduling information;
- raw detector or signal data;
- reconstructed images;
- physiological measurements;
- contrast-agent information;
- acquisition and reconstruction parameters;
- radiation-dose information; and
- reports and other derived clinical information.

2 Data Acquisition

Data acquisition is the process of collecting patient, examination and imaging information from an imaging modality or another healthcare information system. It begins before the first image appears on the monitor.

At the most general level, diagnostic imaging converts a physical interaction into measurable signals and then into numbers that a computer can process.



2.1 Step 1: Interaction with the patient

Each modality interrogates the patient in a different way. The information carried by the resulting signal depends on the underlying physics.

Table 1: Examples of modality-specific signal formation

Modality	Physical interaction or source	What is measured or represented
X-ray	X-ray transmission and differential attenuation	A projection pattern related to how tissues attenuate the beam
CT	X-ray attenuation measured from many projection angles	Projection measurements reconstructed into cross-sectional images
MRI	Nuclear magnetic resonance and radiofrequency signal behaviour	Signals whose contrast depends on sequence design and tissue properties
Ultrasound	Transmission and reflection of acoustic waves	Returning echoes, Doppler information and time-dependent signals
Nuclear medicine	Radiation emitted from administered radiopharmaceuticals	Distribution of tracer activity, often reflecting physiological or metabolic processes

2.2 Step 2: Signal detection

A detector or receiver converts the physical response into an electrical or otherwise measurable signal. Examples include digital X-ray detectors, CT detector arrays, MRI receiver coils, ultrasound transducers and nuclear-medicine detector systems.

The detector is important because it determines how efficiently useful information is captured and how much unwanted variation or noise is introduced.

2.3 Step 3: From analogue information to digital numbers

Many imaging signals begin as continuously varying physical quantities. Computers, however, process information numerically. An **analogue-to-digital converter (ADC)** converts the measured signal into digital values.

Two related ideas are useful:

Sampling

decides *where or when* the signal is measured.

Quantization

decides which numerical level is assigned to the measured signal amplitude.

High-Yield

Digitization is not simply “saving the image to a computer.” It is the conversion of measured physical information into discrete numerical data that can be reconstructed, processed, displayed, transmitted and stored.

2.4 Step 4: Reconstruction and processing

Some modalities create an image quite directly from detector data; others require mathematically intensive reconstruction. CT reconstructs images from many attenuation measurements. MRI reconstructs spatial information from acquired signal data. PET and SPECT use tomographic reconstruction to estimate the spatial distribution of radiotracer activity.

The resulting image can then undergo further processing such as filtering, multiplanar reformatting, windowing, measurement, segmentation or quantitative analysis.

2.5 Step 5: Formatting, communication and storage

Once image data are available, they must be associated with the correct patient and examination information and communicated in a way that other systems understand. This is where DICOM becomes central.

3 Sources and Forms of Imaging Data

The phrase *medical imaging data* should not be interpreted as a single data type. Different modalities produce different combinations of raw data, images, derived maps, measurements and acquisition parameters.

3.1 Digital radiography

Digital radiography produces two-dimensional projection images based on differential X-ray attenuation. Typical examinations include chest, abdomen and extremity radiographs. In addition to pixel data, the stored information can include patient identifiers, examination details and exposure-related parameters.

3.2 Computed tomography

CT generates projection measurements that are reconstructed into cross-sectional images. A CT examination may therefore involve several information levels: raw or projection data, reconstructed image series, reformatted images, acquisition parameters and radiation-dose information.

Example: one CT examination, several datasets

A contrast-enhanced CT of the abdomen may contain an unenhanced series, arterial-phase series, portal-venous series and delayed series. Each series belongs to the same overall examination but may contain hundreds of individual images. Informatics systems must preserve these relationships so that the examination remains organised and interpretable.

3.3 Magnetic resonance imaging

MRI can generate numerous image series from a single examination. Examples include T1-weighted, T2-weighted, FLAIR, diffusion-weighted imaging (DWI), apparent diffusion coefficient (ADC) maps, susceptibility-sensitive imaging, angiographic sequences, perfusion data, spectroscopy and functional MRI.

This illustrates a central informatics problem: the system must preserve not only the images but also the sequence and acquisition information required to understand what each series represents.

3.4 Ultrasound

Ultrasound systems may produce still images, real-time cine loops, spectral Doppler traces, colour Doppler information and measurements. Therefore, a single imaging standard must be flexible enough to represent both static and time-dependent information.

3.5 Nuclear medicine

Nuclear medicine includes planar scintigraphy, SPECT and PET. The data can represent functional, physiological or metabolic processes and may be combined with anatomical CT or MRI information in hybrid imaging systems.

4 DICOM Foundations

DICOM stands for **Digital Imaging and Communications in Medicine**. It is the standard used to represent, communicate and manage medical imaging information across equipment and information systems.

The value of DICOM is interoperability. A scanner and a PACS do not have to come from the same manufacturer merely to exchange imaging information; both can implement the same standardised conventions and services.

DICOM is image format and more

Is the file format used for radiology images. It doesn't stop at that. DICOM also defines how imaging information is structured and how systems communicate imaging-related information.

4.1 What can a DICOM imaging object contain?

An imaging object can contain the actual image pixel data together with descriptive information, often called **metadata**. Examples include:

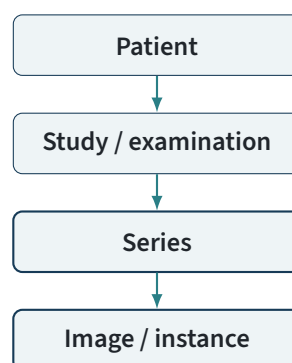
Table 2: Examples of information associated with a DICOM imaging object

Information group	Examples
Image information	Pixel data and image characteristics
Patient information	Patient name, patient identifier, date of birth, sex
Examination information	Modality, study date, study description, accession information
MRI acquisition information	TR, TE, flip angle, slice thickness, field of view, matrix
CT acquisition / dose information	kVp, tube-current information, slice thickness, pitch and dose-related values such as CTDI _{vol} and DLP when available

The critical point is that an image is meaningful clinically only when it remains correctly associated with the information that explains *who* it belongs to, *what* examination was performed and *how* it was acquired.

4.2 A useful organisational model

A practical way to think about imaging information is as a hierarchy:



A patient may have many studies over time. One study may contain several series, and each series may contain many images or other instances. This structure is especially obvious in CT and MRI, where a single examination often contains several acquisition or reconstruction series.

Why correct identifiers matter

If a study is associated with the wrong patient identifier, the problem is not cosmetic. The error can affect retrieval, comparison with previous examinations, reporting and ultimately clinical decision-making. Accurate demographic and examination data are therefore part of patient safety.

5 From Modality to PACS: A Conceptual Workflow

A simplified imaging workflow can be understood as a sequence of information hand-offs:



Later sessions will add the RIS, EHR, storage architecture, security controls, cloud infrastructure and analytics layers to this workflow.

Common Error

Do not treat **PACS**, **RIS** and **DICOM** as interchangeable terms. DICOM is a standard; PACS is principally an image-management system; RIS is principally a radiology workflow and information-management system. Their functions overlap operationally, but the concepts are not synonyms.

6 What the Radiographer Must Get Right

The radiographer is an important source of data quality in the imaging chain. Responsibilities relevant to this session include:

- confirming the correct patient before acquisition;
- ensuring the correct examination is selected;
- checking that demographic and study information are accurate;
- choosing and applying the appropriate imaging protocol;
- recognising when an acquisition or transfer has failed; and
- confirming that images have been successfully sent to the intended imaging system according to local workflow.

Focus

Be able to explain the complete chain in words: **physical interaction** → **detection** → **signal** → **digitization** → **reconstruction** → **DICOM representation** → **communication/storage**. Merely listing these terms is not enough; understand what changes at each stage.

Key Takeaways

- Imaging informatics connects imaging equipment, patient information, image data, healthcare professionals and information systems.
- Data acquisition begins with a physical interaction and ends with usable digital information, not merely with pressing the exposure or scan button.

- Different imaging modalities produce different combinations of raw data, reconstructed images, measurements and acquisition metadata.
- DICOM supports the representation and communication of medical imaging information.
- A DICOM imaging object contains more than pixel values; associated patient, examination and acquisition information provides clinical context.
- Reliable patient identification and correct study information are patient-safety issues as well as informatics issues.

Key Terms

Imaging informatics

Application of information and computing systems to the acquisition, management, communication, analysis and use of medical images.

Data acquisition

Collection of imaging and related information from a modality or healthcare system.

ADC

Analogue-to-digital converter; converts measured analogue information into digital values.

Sampling

Selection of measurement locations or times from a signal.

Quantization

Assignment of measured amplitudes to discrete numerical levels.

DICOM

Digital Imaging and Communications in Medicine; a standard for medical imaging information and communication.

Metadata

Descriptive information associated with the image or imaging object, such as patient, study and acquisition details.

PACS

Picture Archiving and Communication System; introduced here as the destination for image management and developed in Session 2.

Self-Assessment Questions

1. Define imaging informatics and explain why it is necessary in a modern radiology department.
2. Describe the five broad stages from interaction with the patient to digital image storage.
3. Differentiate sampling from quantization.
4. Give two examples of data generated by CT other than the final displayed image.
5. Explain why DICOM should be described as an image-file format and more.
6. List four categories of information that may accompany DICOM image pixel data.

7. Explain the relationship between a patient, study, series and image/instance.
8. A CT image is diagnostically adequate but has been assigned to the wrong patient record. Explain why this is an informatics and patient-safety problem.

Series Navigation

1	Data Acquisition & DICOM Foundations — this session
2	PACS, RIS, Storage Systems & Information Management
3	Cybersecurity in Medical Imaging
4	Cloud-Based Medical Imaging
5	Health Data Analytics, Interoperability & Applied Practice

